MetaSUB: Building a Global Network to Map Urban Microbial Life

*Founded in 2015, the **MetaSUB Consortium** created a worldwide, interdisciplinary network to sample and sequence microbes from transit systems, sewage, hospitals, and other urban environments across more than 100 countries.*

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A history of the MetaSUB consortium: Tracking urban microbes around the globe

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Summary

The MetaSUB Consortium, founded in 2015, is a global consortium with an interdisciplinary team of clinicians, scientists, bioinformaticians, engineers, and designers, with members from more than 100 countries across the globe. This network has continually collected samples from urban and rural sites including subways and transit systems, sewage systems, hospitals, and other environmental sampling. These collections have been ongoing since 2015 and have continued when possible, even throughout the COVID-19 pandemic. The consortium has optimized their workflow for the collection, isolation, and sequencing of DNA and RNA collected from these various sites and processing them for metagenomics analysis, including the identification of SARS-CoV-2 and its variants. Here, the Consortium describes its foundations, and its ongoing work to expand on this network and to focus its scope on the mapping, annotation, and prediction of emerging pathogens, mapping microbial evolution and antibiotic resistance, and the discovery of novel organisms and biosynthetic gene clusters.

Every surface you touched today had microbes on it